

# Ahmed Gamal

AI Engineer | LangGraph / LangChain / Agentic AI Systems  
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## SUMMARY

AI engineering candidate focused on building agentic LLM systems with LangGraph, LangChain, Python, structured outputs, tracing, and evaluation. Previous smart contract auditing background brings strong attention to security, edge cases, tool risk, and failure modes.

## TECHNICAL SKILLS

**Languages:** Python, Solidity

**AI/LLM:** LangGraph, LangChain, LangSmith, AWS Bedrock, structured outputs, tool calling, multi-agent orchestration

**RAG/Retrieval:** PDF/document loading, recursive chunking, SentenceTransformers embeddings, ChromaDB vector storage, semantic retrieval

**AI Evaluation:** Golden datasets, deterministic graders, route/tool-call grading, escalation grading, LLM-as-judge evaluators

**Backend/Data:** FastAPI, async Python, Docker, Pydantic, PostgreSQL, PostgresSaver, pytest, Git, CLI workflows

**AI-Assisted Engineering:** Codex, Claude Code, custom SKILL.md workflows, repo-specific agent instructions, AI-assisted debugging, NotebookLM source-grounded research

**Security:** Smart contract review, Slither, Foundry fuzzing, access control, reentrancy, oracle-risk analysis

## AI ENGINEERING PROJECTS

### Multi-Agent Customer Support Triage System

- Built a LangGraph/LangChain customer-support triage agent using a supervisor-as-tools pattern to route requests across billing, technical, FAQ, and escalation specialists.
- Added Pydantic structured outputs for each specialist result, enabling machine-readable routing confidence, escalation flags, missing-info fields, and source attribution.
- Instrumented the system with LangSmith tags and metadata across root runs, route tools, worker agents, and structured-output parsers for traceable debugging.
- Exposed the LangGraph supervisor through an async FastAPI endpoint with structured request/response models returning answer, route, and escalation state.
- Added Docker packaging for running the FastAPI support-triage service from a reproducible container image.
- Locked file tools to approved documentation and ticket directories, reducing agent risk from arbitrary file reads and writes.
- Built a 20-case eval suite with deterministic graders for route accuracy, escalation behavior, forbidden outputs, content coverage, and hop limits, then added an LLM-as-judge grader to score customer-support answer quality against a rubric.

### LangGraph Research-Writer-Validator Pipeline

- Built a StateGraph-based research pipeline with researcher, writer, and validator nodes connected through explicit graph edges and conditional validation routing.
- Implemented PostgresSaver checkpointing and human-in-the-loop interruption before document generation, allowing review and resume from persisted graph state.
- Used structured validator outputs to approve or request revisions before the graph reached completion.
- Added LangSmith metadata and tags to support trace inspection across the research, writing, and validation workflow.

## SECURITY EXPERIENCE

### Smart Contract Security Auditor | Independent / Competitive Audits | Jan 2025 - Jan 2026

- Reviewed Solidity smart contracts across DeFi, NFT, and staking systems for access-control flaws, reentrancy risk, oracle assumptions, and economic edge cases.
- Used Slither static analysis and Foundry fuzz testing to identify and validate potential vulnerabilities.
- Submitted valid findings in Sherlock and CodeHawks audit contests, including medium- and low-severity issues.

## OTHER EXPERIENCE

### Sales Associate | Alyasra Fashion Company, Kuwait | 2023 - 2024

- Supported retail sales operations, customer service, product guidance, and day-to-day store workflows in a high-volume fashion retail environment.
- Communicated with customers to understand needs, explain product options, and support purchasing decisions.

### Freelance Video Editor | 2024

- Edited short-form video content for freelance clients while managing revisions, delivery deadlines, and client communication.

## EDUCATION

Secondary Education | Jabriya School, Kuwait